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AIR CONDITIONING SYSTEM AND TRANSPORTATION SYSTEM INCLUDING SAME

Abstract

An air conditioning system is provided. The air conditioning system includes an air compressor, a battery, a generator, a blower, and a turboexpander. The air compressor generates pressurized air from ambient air. The battery is for powering the air compressor motor. The generator is electrically connected to the battery. The turboexpander includes a drive shaft, a turbine wheel coupled to the drive shaft, and an expander coupled to the drive shaft. The pressurized air provides a source of kinetic energy to directly cause the turbine wheel to rotate and generate boosted air from the pressurized air, and the boosted air from the turbine wheel is configured to be received in the expander wheel and directly cause the expander wheel and the drive shaft to rotate together with the turbine wheel, thereby generating expanded and cooled air in the expander wheel.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/816,006, filed Aug. 27, 2024, and entitled “AIR CONDITIONING SYSTEM AND TRANSPORTATION SYSTEM INCLUDING SAME”, which is a continuation-in-part of U.S. patent application Ser. No. 18/612,010, filed Mar. 21, 2024, now U.S. Pat. No. 12,103,354, and entitled “AIR CONDITIONING SYSTEM, TRANSPORTATION SYSTEM INCLUDING THE SAME, AND ASSOCIATED METHOD”, which claims priority to and claims the benefit of U.S. Patent Application Ser. No. 63/559,466, filed Feb. 29, 2024, the contents of which are incorporated herein in by reference.

BACKGROUND
[0002] Air conditioning (A/C) systems have been known for many years. Generally, it can be said that A/C systems provide cooled air into an environment (e.g., a residential or commercial building) by removing heat from indoor air. As the A/C system is performing this function, it returns the cooled air to the indoor space, and expels hot air outside of the building. Today, known A/C systems use a coolant, such as freon, ammonia, propane, and the like, and circulates this coolant to generate the cooled air. Furthermore, today's A/C systems, such as the prior art A/C system **2** of FIG. **1**, include a compressor **4**, a condenser **6**, and an evaporator **8**. In operation, these components of the A/C system work to change the coolant from gas to liquid and back to a gas.
[0003] More specifically, the compressor **4** increases the pressure and temperature of the coolant gas and delivers it to the condenser **6**, where it is converted to a liquid, before it is sent back to the evaporator **8**. In the evaporator **8**, the liquid coolant (e.g., freon, ammonia etc.) evaporates and cools the coil of the evaporator **8**. Subsequent to this cooling of the coil of the evaporator **8**, a fan **10** blows air across the cold coil of the evaporator **8** in order to cool the building (e.g., residence, commercial, or otherwise). Additionally, after being blown into the building, the cooled air is then circulated throughout the building while the heated evaporated gas is sent back outside the compressor. In other words, the A/C system **2** is a closed loop system, and as such, the heat is then released into the outdoor air as the coolant returns to a liquid state. In operation, this sequence is repeated until the building reaches a desired temperature.
[0004] The above-described system/process has a number of drawbacks. First, coolant in A/C systems, such as freon (e.g., refrigerant), is becoming more and more regulated. As a result, the coolant is becoming extremely expensive and inefficient for users, who may be homeowners or commercial building owners, to use in their A/C systems. Second, performing maintenance on today's A/C systems is rather difficult. For example, maintenance technicians who have to replace coolant of a given A/C system with new coolant are forced to turn off an entire A/C system, syphon out the old coolant, and introduce the new coolant. This process can take significant time to perform, costing users money. Third, because the cost of various coolants has gotten so high, owners are often forced to replace entire A/C systems when they have leaks in their coils, rather

than simply introduce new coolant. This is not desirable. Fourth, because coolant and today's A/C systems (e.g., evaporators and condensers) are so expensive, many places in the world, such as hot desert regions, are unable to afford the same, thereby making withstanding the heat rather difficult. Fifth because powering air compressors requires users to draw power from a power grid, regions of the world with less sophisticated electric power capabilities are undesirably deprived of air conditioning. Finally, because today's A/C systems are closed loop systems, users are forced to keep their windows and doors closed, in order to prevent cooled air from escaping, and/or to minimize A/C bills.

[0005] Other systems which cause gases to move between states, besides the A/C system **2**, include air separation systems at air separation plants. These systems have been known for a long time. However, the primary purpose of most all if not all air separation systems at these plants is to liquify gases (e.g., nitrogen, oxygen, etc.) for refrigeration purposes. Accordingly, equipment in these plants is uniquely tailored for the purpose of, for example, liquifying oxygen and nitrogen. Furthermore, gases in such plants, such as nitrogen gas, are lethal if breathed in by a human.

[0006] It is with respect to these and other considerations that the instant disclosure is concerned.

SUMMARY

[0007] In one aspect of the disclosed concept, an air conditioning system is provided. The air conditioning system includes an air compressor, a battery, a generator, a blower, and a turboexpander. The air compressor generates pressurized air from ambient air. The battery is for powering the air compressor motor. The generator is electrically connected to the battery. The turboexpander includes a drive shaft, a turbine wheel coupled to the drive shaft, and an expander coupled to the drive shaft. The pressurized air provides a source of kinetic energy to directly cause the turbine wheel to rotate and generate boosted air from the pressurized air, and the boosted air from the turbine wheel is configured to be received in the expander wheel and directly cause the expander wheel and the drive shaft to rotate together with the turbine wheel, thereby generating expanded and cooled air in the expander wheel.

[0008] In another aspect, a transportation system including an electric motor and the aforementioned air conditioning system is provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a schematic of a prior art A/C system.

[0010] FIG. **2** is a simplified view of an A/C system in accordance with the disclosed concept, shown as employed with an electrical apparatus, an additional power source, a vent stack, and ductwork, each of which is illustrated in dashed line drawing, and wherein double line connections between components in FIG. **2** denote electrical connections and single line connections between components in FIG. **2** generally denote conduits (e.g., without limitation, pipes) through which fluids may flow.

[0011] FIGS. **3** and **4** are front and simplified views, respectively, of a transportation system including the A/C system of FIG. **2**, in accordance with one non-limiting embodiment of the disclosed concept.

DETAILED DESCRIPTION

[0012] As employed herein, the term “coupled” shall mean connected together either directly or via one or more intermediate parts or components.

[0013] As employed herein, the term “number” shall mean one or an integer greater than one (i.e., a plurality).

[0014] As employed herein, the term “air compressor” shall mean a device which generates pressurized air from ambient air, and delivers the pressurized air at an outlet at a pressure greater than the pressure of the ambient air entering the inlet. Air compressors in accordance with the disclosed concept may use electrical energy in order to generate a flow of pressurized gas. Air compressors may include an electric air compressor motor. Air compressors in accordance with the disclosed concept, unlike nitrogen compressors (or compressors configured for other gases besides air), may not have gas coolers, inlet and outlet dampers, and asynchronous motors.

[0015] As employed herein, the term “blower” shall mean an apparatus configured to produce air movement to a space. Blowers in accordance with the disclosed concept include propeller fans, centrifugal fans, vane-axial fans, and tube-axial fans.

[0016] As employed herein, the term “generator” shall mean a device configured to convert mechanical energy obtained from an external source into electrical energy as an output.

[0017] Generators in accordance with the disclosed concept convert mechanical energy from rotation of a drive shaft of a turboexpander into electrical energy, and may do so by moving at least one electrical conductor in a magnetic field in order to create a voltage difference between two ends of the electrical conductor.

[0018] As employed herein, the term “coolant” shall mean a substance configured to change states between liquid and gas, and as employed in a coil of an evaporator of an A/C system, such as the A/C system **2**. Non-limiting examples of coolants include freon, propane, and ammonia.

[0019] As used herein, the term “air” shall mean an atmospheric gas comprised of Nitrogen, Oxygen, and Argon. Air in accordance with the disclosed concept includes indoor and outdoor air, as well as both purified and non-purified air, wherein purified air includes, but is not limited to, air in which moisture and/or carbon dioxide have been removed. In one non-limiting example, air in accordance with the disclosed concept is non-toxic (e.g., breathable) to human beings.

[0020] As employed herein, the term “valve” shall mean a device for causing an opening area of a region of a conduit to be changeable as the valve opens and closes. Valves in accordance with disclosed concept include guide vanes, as well as devices which move between FULLY OPEN states and FULLY CLOSED states.

[0021] As employed herein, the terms “inlet” and “outlet” shall each correspond to a “conduit”, whether a metallic or non-metallic pipe, or other type of conduit.

[0022] As employed herein, the term “control system” shall mean a system for directing the flow of air in an air conditioning system by causing any number of valves to independently move to predetermined positions, including fully open positions, fully closed positions, and positions therebetween. Control systems in accordance with the disclosed concept include programmable logic control systems.

[0023] As employed herein, the phrase “combustion reaction involving burning of fuel” shall mean a reaction involving an explosion between a gas, such as air, and a combustible fuel (e.g., oil including byproducts such as gasoline and aviation fuel, and natural gas including byproducts such as hydrogen fuel). A “combustion reaction involving burning of fuel” does not include the discharge of energy from a battery and associated powering of an air compressor motor therewith.

[0024] As employed herein, the phrase “sole source of kinetic energy” shall mean a source of kinetic energy, such as a flow of pressurized air, in whose absence another object, such as a turbine wheel of a turboexpander, is configured to be in a static state (e.g., at rest), with respect to a ground.

[0025] FIG. **2** shows an example A/C system **100**, in accordance with one non-limiting embodiment of the disclosed concept. The A/C system **100** in one example is preferably a coolant-free air conditioning system such that it functions without circulating coolant into a coil of an evaporator or other device. This is highly beneficial, as compared to the A/C system **2** (FIG. **1**), which is required to use coolant to cool buildings and the like. More specifically, by being coolant-free, the A/C system **100** can, among other advantages, save users money and minimize maintenance. That is, large evaporators and the like do not have to be replaced, and expensive coolant does not have to be employed in order to cool a building.

[0026] The A/C system **100** will now be discussed in detail, and the aforementioned advantages as well as others will be made apparent. As shown in FIG. **2**, the system **100** includes an air compressor **110**, a turboexpander **120**, a generator **140** coupled to the turboexpander **120**, a battery **145** electrically connected to the generator **140**, and a blower **150**. The turboexpander **120** is coupled to the air compressor **110**, the generator **140**, and the blower **150**. In one non-limiting example, the A/C system **100** is configured to deliver expanded and cooled air to ductwork **160**, which may be coupled to the blower **150** and configured to deliver the expanded and cooled air from the turboexpander **120** throughout a building.

[0027] In operation, aspects of the A/C system **100** comprising the air compressor **110**, the turboexpander **120**, the generator **140**, the battery **145**, and the blower **150** may be located outside of a building, and may be fluidly coupled to the ductwork **160**, which may extend throughout the interior of the building. In a suitable alternative example, the air compressor **110**, the turboexpander **120**, the generator **140**, and the blower **150** may all be located in an interior of a building and be a self-contained subassembly that is configured to pull indoor air into the air compressor **110** for cooling, wherein the cooled air may be delivered back into the indoor environment.

[0028] The turboexpander **120** includes a drive shaft **122**, a turbine wheel **124** coupled to the drive shaft **122**, an expander wheel **126** coupled to the drive shaft **122** and the turbine wheel **124** (e.g., via the drive shaft **122**), a first inlet **128** and a first outlet **130** each associated with the turbine wheel **124**, and a second inlet **132** and a second outlet **134** each associated with the expander wheel **126**. Furthermore, as shown, the generator **140** includes a rotor (“R”) **142** coupled to the drive shaft **122** of the turboexpander **120** and configured to be rotated during operation of the turboexpander **120**.

[0029] Moreover, the air compressor **110** may include an air compressor motor **119**, which may be an electric motor. Additionally, the generator **140** is preferably electrically connected to the air compressor motor **119** in order to power the air compressor **110**, and/or is configured to be electrically connected to an electrical apparatus **180** (shown in dashed line drawing in FIG. **2**) in order to power the electrical apparatus **180**. The electrical apparatus **180** may be an operating system of a building (e.g., a restaurant, hospital, and the like), such as a lighting system, refrigeration system, freezer system, etc., in a manner wherein the A/C system **100** is configured to deliver air conditioning to the building, and also supply power to the lighting, refrigeration, and/or freezer system, as well as various other systems within the building. Furthermore, the generator **140** may be configured to power, e.g., fully power, the electrical apparatus **180** (e.g., the operating system of the building) while the blower **150** is delivering the expanded and cooled air to the building. After discussion of the operation of the A/C system **100**, below will be discussion of an example implementation of the A/C system **100**, wherein an electrical apparatus is in the form of an electric motor **280** (FIG. **4**) of a transportation system (e.g., vehicle **200**, shown in FIGS. **3** and **4**), such that the vehicle **200** includes the A/C system **100** as an integral operating system.

[0030] Continuing to refer to FIG. **2**, the battery **145** is electrically connected to the air compressor motor **119**, the generator **140**, and the electrical apparatus **180**. Additionally, in non-limiting examples of the disclosed concept, the A/C system **100** does have the capability to receive power from a grid (e.g., from a power source **145-1**, shown in dashed line drawing in FIG. **2**), should the need arise.

[0031] In operation, the generator **140** reliably supplies power to a number of devices. For example, the battery **145** is electrically connected to and configured to be charged by the generator **140** of the A/C system **100**. Moreover, the battery **145** is also configured to power the air compressor motor **119**. Specifically, during operation of the A/C system **100**, power may first be drawn by the air compressor motor **119** from the battery **145**.

[0032] In one non-limiting example, the A/C system **100** further includes a moisture separator (“MS”) **114**, which may be a filter, fluidly coupled to an outlet **113** of the air compressor **110** and the first inlet **128** of the turboexpander **120** in order for moisture (e.g., water) to be removed from the pressurized air before the pressurized air is delivered to the turboexpander **120**. As such, the air compressor **110** may be configured to pull in ambient air at an inlet **111** of the air compressor **110** from the atmosphere, generate pressurized air from the ambient air, and cause the pressurized air to exit via the outlet **113** before being passed through the moisture separator **114**. It will be appreciated that the ambient air may be pressurized to any number of different pressure ranges, depending on the environment for the A/C system **100**. In one example, the ambient air is pressurized to between 130-180 psi. In an alternative example, an air compressor is configured to generate higher or lower pressures higher than 130-180 psi, provided suitable materials are employed, such as thicker and higher-grade steels for higher pressures.

[0033] Continuing to refer to FIG. **2**, it will be appreciated that the air compressor **110** may, in one example, be fluidly coupled to the turboexpander **120** without any intermediate air processing apparatuses (e.g., a heat exchanger, distillation column, etc.) therebetween, such that the pressurized air flows directly from the air compressor **110** into the turboexpander **120**. As used herein, the moisture separator **114** is not considered an air processing apparatus. In a further example, the outlet **113** of the air compressor **110** is directly coupled to and engaged with the first inlet **128**, such that air flows directly from the outlet **113** through the moisture separator **114** and into the first inlet **128** of the turboexpander **120**, without passing through any other intermediate parts or components. It will also be appreciated that the blower **150** may be fluidly coupled to the second outlet **134** without any intermediate air processing apparatuses (e.g., a heat exchanger, distillation column, etc.) being located therebetween. In other words, the blower **150** may have an inlet **137** that is directly coupled with and engaged with the second outlet **134**.

[0034] In one example, a suitable alternative air compressor (not shown) may be fluidly coupled to the first inlet **128** of the turboexpander via a side stream **115**, shown in FIG. **2**. In such an embodiment, the alternative air compressor (not shown) may be an existing air compressor at a garage, air separation facility, or the like. In such an example embodiment of the disclosed concept, the side stream **115**, and a valve **115-1** coupled to the side stream **115**, may be one of a plurality of streams of the air compressor (not shown), wherein other of the streams may separately be configured to be directed to, for example, a fork-lift (not shown) in a garage or other location. Accordingly, it will be appreciated that air from the alternative air compressor (not shown) may likewise not be processed (e.g., in a heat exchanger, distillation column, or the like) when passing through the valve **115-1** of the side stream **115** to the first inlet **128**. Furthermore, in such an example of the disclosed concept, an air conditioning system preferably includes the turboexpander **120**, the generator **140**, the battery **145**, and the blower **150**, which are configured to receive the pressurized air from the separate air compressor (not shown) via the side stream **115** and valve **115-1**.

[0035] Continuing to refer to FIG. **2**, coupled to the outlet **113** of the air compressor **110** is a valve **113-1**, and coupled to the first inlet **128** of the turboexpander **120** is another valve **128-1**. In operation, the valves **113-1**,**128-1** function to control the volume and speed of air both exiting the air compressor **110** and entering the first inlet **128** of the turboexpander **120**. As a result, volume flow of air moving through the A/C system **100** is configured to be reliably controlled by opening and closing the valves **113-1**,**128-1**.

[0036] More specifically, the A/C system **100** may further include a control system **190**. In one example, the control system **190** is wirelessly connected and/or electrically connected to the air compressor **110** and the valves **113-1**,**128-1**, such that responsive to actuation of the control system **190** (e.g., wherein a user presses a button on a control panel or sends a signal to the control system **190** with a wireless device), the air compressor **110** is configured to generate the pressurized air at a predetermined first pressure, thereby causing the turboexpander **120** to generate expanded and cooled air at a predetermined second temperature. This is achieved by the valves **113-1**,**128-1** being opened and closed to predetermined opening areas by the control system **190**.

[0037] Once inside the turboexpander **120**, the turbine wheel **124** may generate boosted air from the pressurized air from the air compressor **110**, and the expander wheel **126** may generate expanded and cooled air from the boosted air, as will be discussed below. For example, the turbine wheel **124** boosts the pressurized air from the air compressor **110** to an even greater pressure. As such, by employing the turbine wheel **124**, a size of the air compressor **110** is advantageously able to be relatively small. In one example, the turbine wheel **124** receives the pressurized ambient air at 130-180 psi and boosts the pressurized ambient air to a pressure even greater than 180 psi. After being boosted by the turbine wheel **124**, the even higher pressurized air exits the turbine wheel **124** via the first outlet **130**, which is fluidly coupled to the second inlet **132** of the expander wheel **126**.

[0038] Accordingly, after exiting through the first outlet **130**, the boosted air enters the expander wheel **126** via the second inlet **132**. Once inside the expander wheel **126**, the boosted air is expanded by volume, which causes a relatively large pressure and temperature drop in the air. In other words, the expander wheel **126** generates expanded and cooled air from the boosted air. In one example, the air is configured to exit the turboexpander at between 3-20 psi of pressure, as well as between 33-45 degrees Fahrenheit. It will be appreciated that inlet and outlet temperatures associated with the turboexpander **120** may readily be varied.

[0039] However, in accordance with the disclosed concept, air is preferably not being liquified during operation of the A/C system **100** (e.g., other than at the moisture separator **114**), and operation of the A/C system **100** includes directing the air above a freezing temperature of water. That is, the air compressor **110** and the turboexpander **120** are together configured to change the ambient air to the expanded and cooled air without causing any of the ambient, pressurized, boosted, and expanded and cooled air to change states between a gas and a liquid. This includes embodiments where air undergoes a purification process corresponding to moisture and carbon dioxide being removed from the air between the air compressor **110** and the first inlet **128**.

[0040] Furthermore, it will be appreciated that the gas which is compressed by the air compressor **110** and the turbine wheel **124**, and then expanded and cooled by the expander wheel **126**, is the same gas which is delivered to the blower **150**. That is, unlike existing systems (e.g., the A/C system **2**) in which coolant changes from liquid to gas in the evaporator in order to cool an entirely separate and distinct gas (e.g., air) which is passed over the coil of the evaporator, the A/C system **100** is preferably coolant-free. As such, the same air which is pressurized, boosted, and expanded and cooled is also the air that is delivered to the blower **150**, such that the blower receives the expanded and cooled air in order to deliver the expanded and cooled air to an environment. Put another way, substantially all of the air associated with the A/C system **100** enters and passes through each of the compressor **110**, the turboexpander **120**, and the blower **150** such that coolant (e.g., being located in coils) is preferably not employed to cool the air. As such, the A/C system **100** readily provides tremendous advantages over the prior art A/C system **2** (FIG. **1**) and improves over the prior art A/C system **2** in most and/or all of the respects discussed above in connection with the A/C system **2** (FIG. **1**).

[0041] Continuing to refer to FIG. **2**, as the expander wheel **126** is dropping the pressure and temperature of the air, the drive shaft **122** is rotating, which in turn powers the windings of the generator **140** via the coupling between the rotor **142** and the drive shaft **122**. The purpose of this functionality will be described below. Additionally, once the pressure and temperature of the air have been dropped by the expander wheel **126**, the expanded and cooled (e.g., and also low pressure) air is caused to exit the turboexpander **120** via the second outlet **134**. In turn, this cool and low-pressure air is advantageously used to cool a building, the cabin of the vehicle **200** (FIGS. **3** and **4**), and/or other environments. More specifically, and continuing to refer to FIG. **2**, the inlet **137** of the blower **150** is fluidly coupled to the second outlet **134**. As such, when the cool and low-pressure air is received at the blower **150**, a fan of the blower **150** can cause that cool and low-pressure air to be excited (e.g., made more turbulent), thereby allowing it to be delivered more effectively to a building.

[0042] Furthermore, because the air is rather cold, in one example the A/C system **100** is configured to be employed with a valve system in the form of a vent stack **152** coupled to the outlet of the blower **150**. The vent stack **152** may open and close to an outside environment automatically or from a user. In one example, the control system **190** is wirelessly connected to the vent stack **152** in order to better control the temperature of cooled air being delivered to a building. For example, a smaller opening of the vent stack **152** via the control system **190** is configured cause colder temperatures in a given building. Accordingly, the vent stack **152** may be a way for the A/C system to regulate temperature in a building. In one example, the vent stack **152** could be open one at an initial time one hundred percent and then can close in a gradual manner in order to efficiently regulate temperature in a building.

[0043] Continuing to refer to FIG. **2**, the ductwork **160**, which is coupled to the A/C system **100**, may include a primary branch **162** and a plurality of secondary branches **164**,**166**,**168** fluidly coupled to the primary branch **162**. Additionally, coupled to, located on, and/or provided with an end of each of the secondary branches **164**,**166**,**168** may be a corresponding vent **165**,**167**,**169** (e.g., one which may be positioned adjacent a wall of a room of a commercial building or house). As such, it will be appreciated that the cool and low-pressure air, which has been excited by the blower **150**, will enter the primary branch **162** (e.g., the primary branch **162** is fluidly coupled to the blower **150**) and be forced through each of the secondary branches **164**,**166**,**168**, where it will exit the ductwork **160** through the corresponding vents **165**,**167**,**169**, thereby adequately delivering the cooled and low-pressure air throughout a large number of regions of a building.

[0044] In addition to delivering cooled and low-pressure air to an environment, the A/C system **100** is also provided with a power generation capability in tandem with the air conditioning capability. More specifically, as stated above, as the drive shaft **122** of the turboexpander **120** is caused to rotate, the rotor **142** of the generator **140** is rotated, thus generating power. As shown in FIG. **2**, a first electrical line **170** is electrically connected to the generator **140**, and a number of branch electrical lines **172**,**174**,**176** are each directly or indirectly electrically connected to the first electrical line **170**. In one example, the electrical lines **170**,**172**,**174**,**176** allow electrical power from the generator **140** to be directed to any combination or all of the air compressor motor **119**, the battery **145**, and the electrical apparatus **180**.

[0045] In one example, electric power from the generator **140** is directed back into the air compressor motor **119** via the electrical lines **170**,**172**, which may be one single electrical line. In another example, the A/C system **100** further is configured to deliver power to the electrical apparatus **180**, wherein electric power from the generator **140** is directed to the electrical apparatus **180** via the electrical lines **170**,**174**, which may be one single electrical line. In yet a further example, electric power from the generator **140** is directed to the battery **145** via the electrical lines **170**,**172**,**176**, which may be one single electrical line, in order that the battery **145** may be charged by the generator **140**. Accordingly, the A/C system **100**, unlike the A/C system **2** (FIG. **1**), is advantageously configured to deliver cool and low-pressure air to an environment and simultaneously (e.g., at the same time) deliver electric power to electrical apparatuses, including the air compressor motor **119**, the battery **145**, and/or the electrical apparatus **180**.

[0046] Continuing to refer to FIG. **2**, the air compressor **110** includes the inlet **111** for receiving the ambient air, and the inlet **137** of the blower is configured to receive the expanded and cooled air from the second outlet **134** of the turboexpander **120**. Furthermore, as shown in FIG. **2**, the A/C system **100** further includes a recirculation conduit **139** fluidly coupled to each of the second outlet **134** of the turboexpander **120**, the inlet **137** of the blower **150**, and the inlet **111** of the air compressor **110** in order to recirculate a first amount of the expanded and cooled air back into the inlet **111** of the air compressor after the first amount has exited the turboexpander **120**. In one example, the air compressor **110** is fluidly coupled between the recirculation conduit **139** and the turbine wheel **124** such that the first amount is configured to flow directly from the recirculation conduit **139** into the inlet **111** of the air compressor **110** to be pressurized before flowing into and being boosted by the turbine wheel **124**. In other words, the recirculation conduit **139** is preferably spaced from the inlet **128** associated with the turbine wheel **124**.

[0047] To illustrate, and with continued reference to FIG. **2**, the A/C system **100** may further include a vent conduit **138**, as well as a plurality of valves **111-1**,**137-1**, **138-1**,**139-1** each coupled to a corresponding one of the inlet **111** of the air compressor **110**, the inlet **137** of the blower **150**, the vent conduit **138**, and the recirculation conduit **139**. The valves **111-1**,**137-1**, **138-1**,**139-1** are preferably each wirelessly connected and/or electrically connected to the control system **190**, and function to control the flow of the expanded and cooled air in the system. For example, responsive to actuation of the control system **190**, the valve **138-1** may move between fully open and fully closed states, thus causing either large amounts or none, respectively, of the expanded and cooled air to exit the A/C system to an atmosphere via the vent conduit **138**. Similarly, the valves **137-1**,**139-1**,**111-1** control airflow into the blower **150**, through the recirculation conduit **139**, and into the air compressor **110**.

[0048] In one example, the valves **137-1**,**139-1** function to control the first amount of the expanded and cooled air flowing into the inlet **111** of the air compressor **110** and a second amount of the expanded and cooled air flowing into the blower **150**. In one example, the vent conduit **138** is fluidly coupled to each of the second outlet **134** of the turboexpander **120**, the inlet **137** of the blower **150**, and the recirculation conduit **139**, such that the vent conduit **138** is configured to vent a second amount of the expanded and cooled air to an atmosphere. It will thus be appreciated that the valve **138-1** is configured to control the second amount of the expanded and cooled air being vented to the atmosphere. In one example, responsive to actuation of the control system **190**, such as a user pressing a button on the control system **190** or the control system **190** receiving a signal from a smart device (e.g., mobile phone, tablet, vehicle, etc.), the amount of the expanded and cooled air passing through the recirculation conduit **139** may be a predetermined amount.

[0049] Continuing to refer to FIG. **2**, the air compressor is further configured to startup in a reliable manner. More specifically, the air compressor **110** preferably further includes a vent conduit **117**, the outlet **113**, a valve **117-1** coupled to the vent conduit **117**, and the valve **113-1** coupled to the outlet **113** of the air compressor **110**. Furthermore, the vent conduit **117** is configured to vent an amount of the pressurized air to an atmosphere, and the valve **117-1** is configured to control the amount of the pressurized air being vented to the atmosphere. Moreover, the valve **113-1** is configured to control an amount of the pressurized air exiting the air compressor **110** before being delivered to the turboexpander **120**. In turn, responsive to the A/C system **100** moving from an INITIAL STATE to an OPERATING state, the amount of pressurized air vented to the atmosphere decreases as the amount of air passing through the outlet **113** increases. Put another way, the vent conduit **117** may close gradually or entirely after the air compressor **110** gets going, in order to allow for maximum delivery of the pressurized air to the turboexpander **120**.

[0050] In another example embodiment of the disclosed concept, and as shown in FIGS. **3** and **4**, the vehicle **200** includes the A/C system **100**, a vehicle body or frame **203**, a pair of wheels **204**, at least one element (e.g., first and second axles **207**,**209**) each coupled to the frame **203** as well as to one of the pair of wheels **204**, and an electrical apparatus in the form of the electric motor **280**. In operation the electric motor **280** is configured to drive (e.g., cause to rotate) at least one of the first and second axles **207**,**209**, thereby allowing the first, second, third and fourth wheels **204** to roll when the vehicle **200** is in an OPERATING state. Furthermore, the electric motor **280** is advantageously configured to draw power from each of the generator **140** (FIG. **2**) and the battery **145** (FIG. **2**).

[0051] Continuing to refer to FIG. **4**, it will be appreciated that the generator **140** (FIG. **2**) and the battery **145** of the A/C system **100** are each electrically connected to and configured to supply power to the electric motor **280**. Accordingly, operation of the A/C system **100** as discussed above is configured to deliver cool air to an environment (e.g., a cabin of the vehicle **200**) and also generate electric power in the generator **140** in order to charge the battery **145** as well as power the electric motor **280**. As the electric motor **280** is powered by the generator **140** and/or the battery

145 (FIG. **2**), the body or frame **203** and associated wheels **204** can thus be moved along a road, parking lot, or elsewhere. In other words, the vehicle **200** can be powered and caused to drive by the power from the generator **140** being delivered to the electric motor **280**.

[0052] Additionally, in the example of FIG. **4** the battery **145** for powering the air compressor motor **119** may be configured to supply power to the electric motor **280** in the form of a backup power supply to the electric motor **280** and/or in order to start the electric motor **280**. Accordingly, the generator **140** is configured to power the electric motor **280** and function as a power source during operation of the vehicle **200**. As such, the vehicle **200** may be powered with either relatively little electric charging and/or filling up of a gas tank (for hybrid vehicles). That is, the generator **140** may significantly supplement the power capabilities of the vehicle **200**.

[0053] More specifically, the vehicle **200** may have an OFF state corresponding to the vehicle **200** being turned off and not moving, an IDLING state corresponding to the vehicle **200** being turned on and not moving, and an OPERATING state corresponding to the vehicle **200** being turned on and moving. In one example, the electric motor **280** is configured to draw a first amount of power from the battery **145** (FIG. **2**) in order to move the vehicle **200** from the OFF state to the IDLING state. That is, the battery **145** can get the vehicle **200** started, and then once the vehicle **200** is running, the generator **140** (FIG. **2**) can charge the battery **145** (FIG. **2**) as well as power the electric motor **280**.

[0054] For example, when the vehicle **200** is in the OPERATING state, the electric motor **280** is configured to draw a second amount of power from the generator **140** (FIG. **2**). In other words, the electric motor **280** may draw power from each of the battery **145** and the generator **140**, and the generator **140** may be configured to power and charge the battery **145** during operation of the A/C system **100**. It will also be appreciated that when the vehicle **200** moves from the IDLING state to the OPERATING state, the first amount of power from the battery **145** decreases as the second amount of power from the generator **140** increases.

[0055] Accordingly, the vehicle **200** is configured such that the battery **145** may be employed to start the vehicle **200**, and during operation of the vehicle **200**, more power may be drawn from the generator **140** than the battery **145**. As such, when the vehicle **200** is in the OPERATING state, the first amount of power from the battery **145** is configured to be less than the second amount of power from the generator **140**. As a result, the battery **145** may be substantially smaller than batteries associated with today's electric and hybrid vehicles, wherein these batteries (not shown) are typically required to be relatively large and expensive because they are mostly the sole power sources for the vehicles.

[0056] Furthermore, by employing the A/C system **100** in the vehicle **200**, a tremendous cost and material savings to businesses and consumers can be realized. More specifically, known transportation systems which employ relatively large batteries can advantageously be provided with the relatively small battery **145** instead, due to the generator **140** harnessing energy from the air. As a result, the cost of the vehicle **200** can be relatively low, and the vehicle **200** can also operate in a clean-energy manner wherein material use associated with batteries can be largely minimized.

[0057] More specifically, in one example the air compressor **110** is configured to pull in ambient air having a first amount X**1** of energy, the air compressor **110** is configured to add a second amount X**2** of energy to the ambient air that is being pressurized therein via compression, and during expansion of the air in the expander wheel **126**, energy is removed from the air, such that the expanded and cooled air may be generated for air conditioning. Simultaneously, the expander wheel **126** is configured to create mechanical work in the generator **140** in order to generate electricity for powering the air compressor **110** and/or an electrical apparatus (e.g., the electrical apparatus **180** or the electric motor **280**). As a result, apart from any friction associated losses in the system, a first fraction of the (X**1**+X**2**) energy may be provided as the expanded and cooled air for air conditioning, and a second fraction, which may be a larger fraction, of the (X**1**+X**2**) energy may be routed back from the generator **140** in the form of electricity to the air compressor motor **119**, the battery **145**, and/or the electric motor **280**.

[0058] Additionally, because warmer air has more energy than colder air, the air conditioning system **100** preferably further has a heating element (e.g., heating coil **139-2**) coupled to the recirculation conduit **139**. This is optimal for cold weather conditions. More specifically, the heating coil **139-2** is preferably electrically connected to the generator **140** via the electrical line **178** and is preferably configured to be powered by the generator **140** during operation of the air conditioning system **100**. As a result, the vehicle **200** is optimally configured to be powered by the air conditioning system **100** in cold weather, where the energy from the air is relatively low, as well as warm weather, where the energy from the air is relatively high. That is, the expanded and cooled air entering the recirculation conduit **139** after exiting the second outlet **134** may be heated with the heating coil **139-2** as it is passed through the recirculation conduit **139**, thus increasing an amount of energy supplied to the turboexpander **120** from the air compressor **110**.

[0059] Additionally, as shown in FIG. **4**, the battery **145** (FIG. **2**) of the A/C system **100** may, if the need arises, also be separately charged by connecting the vehicle **200** to a power source **250**, which may be a charging station or plug extending from a building to a charging port of the vehicle **200**.

[0060] It will thus be appreciated that the electric motor **280** may be configured to consume a first amount of power per mile during use, and the generator **140** (FIG. **2**) of A/C system **100** may be configured to deliver a relatively large fraction of the first amount of power to the electric motor **280**. In one example, the first amount of power consumed by the electric motor **280** is between 150-400 watt-hours per mile (e.g., 0.15-0.4 kWh/mile) during use.

[0061] Additionally, by powering the electric motor **280** with the air conditioning system **100**, operation of the vehicle **200** can be said to be safer than known methods of powering a vehicle. For example, accidents involving gasoline powered vehicles may result in fuel tanks exploding, accidents involving vehicles powered by relatively large batteries may result in non-dry portions of the relatively large batteries igniting, and more recent but known technologies involving liquid hydrogen to power a vehicle may be dangerous in that liquid hydrogen is often rather difficult to contain and work with. In accordance with the disclosed concept, by way of contrast, the air conditioning system **100** obviates these challenges by powering the vehicle **200**, in one example, without an explosion between air (e.g., any of the ambient, pressurized, boosted, and expanded and cooled air) and other gases, liquids, or solids. Rather, the disclosed concept relies on a safe heat exchange between energy in the air and electricity.

[0062] Although the disclosed concept has been described in association with the transportation system being in the form of the vehicle **200**, which is an automobile, it will be appreciated that suitable alternative transportation systems are within the scope of the disclosed concept, including non-sedan automobiles such as sports-utility-vehicles and trucks, as well as buses, trains, large or small ships, and the like, wherein each of these transportation systems may have an electric motor electrically connected to the A/C system **100**, and elements coupled to and configured to be driven by the electric motor.

[0063] Accordingly, it will be appreciated that one method of delivering expanded and cooled air to an environment with the A/C system comprises the steps of pulling in ambient air and generating pressurized air from the ambient air, with the air compressor **110**; delivering the pressurized air to a turboexpander **120** of the A/C system **100**; generating boosted air with the turboexpander from the pressurized air from the air compressor **110**; after the boosted air has been generated by the turboexpander **120**, generating the expanded and cooled air from the boosted air with the turboexpander **120**, and also creating mechanical work in a generator **140** of the air conditioning system **100** which is coupled to the turboexpander **120**, in order to generate electricity for powering the air compressor **110** and/or an electrical apparatus **180**; and at the blower **150**, receiving the expanded and cooled air from the turboexpander **120** and delivering the expanded and cooled air to the environment.

[0064] It will also be appreciated that the method may further comprise providing with the air compressor **110** the inlet **111** for receiving the ambient air; providing with the turboexpander **120** the outlet **134** through which the expanded and cooled air exits the turboexpander **120**; providing with the blower **150** an inlet for receiving the expanded and cooled air from the outlet **134** of the turboexpander **120**; and providing the A/C system with a recirculation conduit **139** fluidly coupled to each of the outlet **134** of the turboexpander **120**, the inlet **137** of the blower **150**, and the inlet **111** of the air compressor **110**; and recirculating a first amount of the expanded and cooled air back into the inlet **111** of the air compressor **110** after the first amount has exited the turboexpander **120**. The method may further include powering the operating system (e.g., the electrical apparatus **180**) of a building with the generator **140** while the blower **150** is delivering the expanded and cooled air to the interior of the building.
[0065] Referring again to FIG. **2**, and also FIGS. **3** and **4** which include the A/C system **100**, additional operation of the A/C system **100** will now be described. In one example, it will be appreciated that the turbine wheel **124** is coupled to the drive shaft **122** and is fluidly coupled to the air compressor **110** such that the pressurized air from the air compressor **110** is configured to be forced through a conduit **112** (e.g., wherein the conduit **112** includes the outlet **113** and the inlet **128**, and may be a pipe, such as a closed pipe or continuous closed pipe, which has one single lumen extending from a first open end to a second open end) extending between the air compressor **110** and the turbine wheel **124**, and into the turbine wheel **124**, thereby providing a source of kinetic energy, optionally a sole source of kinetic energy, to directly cause the turbine wheel **124** to rotate and generate the boosted air from the pressurized air.
[0066] The conduit **112** is different than arrangements of known technologies in that it allows for the pressurized air to provide this source of kinetic energy. In accordance with the disclosed concept, the conduit **112** advantageously is configured to receive at least **98** percent of the pressurized air therethrough for delivery to the turbine wheel **124**. Additionally, the air compressor **110** may be configured to discharge the pressurized air at a first pressure, and the pressurized air is preferably configured to be received at the turbine wheel **124** at a second pressure equal to at least **95** percent of the first pressure (e.g., due to the conduit **112**).
[0067] Furthermore, and continuing to refer to FIG. **2**, the expander wheel **126** is preferably coupled to the drive shaft **122** and fluidly coupled to both the turbine wheel **124** and the blower **150** such that the boosted air from the turbine wheel **124** is configured to be received in the expander wheel **126** and directly cause the expander wheel **126** and the drive shaft **122** to rotate together with the turbine wheel **124** (e.g., the drive shaft **122** and the wheels **124**,**126** all rotate at the same angular velocity), thereby generating expanded and cooled air in the expander wheel **126**.
[0068] It will also be appreciated that the air conditioning system is configured such that the expander wheel **126** and the drive shaft **122** rotate together with the turbine wheel **124** via the pressurized air, and the mechanical work is created in the generator **140**, without the turboexpander **120** being driven by a combustion reaction involving burning of fuel by an apparatus directly connected to the turboexpander **120** or by an apparatus indirectly connected thereto via intermediate components. For example, the pressurized air is configured to provide the source of kinetic energy to directly cause the turbine wheel **124** to rotate and generate the boosted air from the pressurized air without the pressurized air being mixed with fuel from a combustion reaction before being received in the turboexpander **120**. As a result, it is contemplated that all of the pressurized, boosted, and expanded and cooled air are non-toxic to humans and are thus breathable. This is distinct from many known technologies employing turboexpanders, such as in airplanes, in which air is mixed with fuel for directly or indirectly driving the turboexpander. Additionally, it will be appreciated that the turboexpander **120** of the air conditioning system **100** is preferably not configured to be driven by any gases other than air.
[0069] Additionally, as stated above, the expander wheel **126** is configured to create mechanical work in the generator **140** responsive to rotation of the drive shaft **122** in order to generate

electricity for charging the battery **145** and thereby powering at least one of the air compressor motor **119** and a separate electrical apparatus (e.g., the electric motor **280**). Thus, the battery **145** is configured to be charged by recycled energy from both the battery **145** and the ambient air after the ambient air has been pressurized by the air compressor **110** and boosted by the turbine wheel **124**.
[0070] Additionally, in one example the A/C system **100** includes desirable apparatus to allow the A/C system **100** to operate at relatively low temperatures. These apparatus include the moisture separator **114**, discussed above, and a carbon dioxide removal system ("CO2 RS") **116**. The moisture separator **114** is fluidly coupled to the conduit **112** and is configured to remove moisture from the pressurized air before the pressurized air is delivered to the turbine wheel **124**, thereby allowing the A/C system **100** to generate the expanded and cooled air at a temperature below a freezing temperature of water. Moreover, the carbon dioxide removal system **116** is preferably coupled to the conduit **112** after the moisture separator **114** and is configured to remove carbon dioxide from the pressurized air after the moisture has been removed by the moisture separator **114**, and before the pressurized air is delivered to the turbine wheel **124**, thereby allowing the A/C system **100** to generate the expanded and cooled air at another temperature below a freezing temperature of carbon dioxide. Because the moisture separator **114** and the carbon dioxide removal system **116** are uniquely configured to allow the expanded and cooled air to exit the expander wheel **126** at low temperatures, these two elements also advantageously allow the A/C system **100** to operate with a relatively large pressure drop across the expander wheel **126**.
[0071] Continuing to refer to FIG. **2**, it will be appreciated that the A/C system **100** further includes an additional conduit **131**, one including the outlet **130** associated with the turbine wheel **124** and the inlet **132** associated with the expander wheel **126**, such that the conduit **131**, which may be a pipe, preferably has a first open end at the turbine wheel **124**, a second open end at the expander wheel **126**, and a closed portion extending therebetween such that the pipe has one single lumen extending from at the turbine wheel **124** to at the expander wheel **126**. As a result of the conduit **131**, the boosted air is allowed to flow directly from the turbine wheel **124** into the expander wheel **130**.
[0072] While the present disclosure has been described with reference to various implementations, it will be understood that these implementations are illustrative and that the scope of the disclosure is not limited to them. Many variations, modifications, additions, and improvements are possible. More generally, implementations in accordance with the present disclosure have been described in the context of particular implementations. Functionality can be separated or combined in blocks differently in various implementations of the disclosure or described with different terminology. These and other variations, modifications, additions, and improvements can fall within the scope of the disclosure as defined in the claims that follow.

Claims

1. An air conditioning system, comprising: an air compressor comprising an air compressor motor configured to pull in ambient air and generate pressurized air from the ambient air; a battery for powering the air compressor motor; a generator electrically connected to the battery; a blower; and a turboexpander comprising: a drive shaft coupled to the generator, a turbine wheel coupled to the drive shaft and fluidly coupled to the air compressor such that the pressurized air from the air compressor is configured to be forced through a conduit extending between the air compressor and the turbine wheel, and into the turbine wheel, thereby providing a source of kinetic energy to directly cause the turbine wheel to rotate and generate boosted air from the pressurized air, and an expander wheel coupled to the drive shaft and fluidly coupled to both the turbine wheel and the blower such that the boosted air from the turbine wheel is configured to be received in the expander wheel and directly cause the expander wheel and the drive shaft to rotate together with the turbine wheel, thereby generating expanded and cooled air in the expander wheel, wherein the expander wheel is configured to create mechanical work in the generator responsive to rotation of the drive shaft in order to generate electricity for charging the battery and thereby powering at least one of the air compressor motor and a separate electrical apparatus, wherein the blower is configured to receive the expanded and cooled air from the expander wheel and deliver the expanded and cooled air to an environment.

2. A transportation system, comprising: an electric motor; at least one element coupled to and configured to be driven by the electric motor in order to move the transportation system between an IDLING state corresponding to the transportation system being turned on and not moving, and an OPERATING state corresponding to the transportation system being turned on and moving; and an air conditioning system, comprising: an air compressor comprising an air compressor motor configured to pull in ambient air and generate pressurized air from the ambient air; a battery for powering the air compressor motor and the electric motor; a generator electrically connected to the battery; a blower; and a turboexpander comprising: a drive shaft coupled to the generator, a turbine wheel coupled to the drive shaft and fluidly coupled to the air compressor such that the pressurized air from the air compressor is configured to be forced through a conduit extending between the air compressor and the turbine wheel, and into the turbine wheel, thereby providing a source of kinetic energy to directly cause the turbine wheel to rotate and generate boosted air from the pressurized air, and an expander wheel coupled to the drive shaft and fluidly coupled to both the turbine wheel and the blower such that the boosted air from the turbine wheel is configured to be received in the expander wheel and directly cause the expander wheel and the drive shaft to rotate together with the turbine wheel, thereby generating expanded and cooled air in the expander wheel, wherein the expander wheel is configured to create mechanical work in the generator responsive to rotation of the drive shaft in order to generate electricity for charging the battery and thereby powering the air compressor motor and the electric motor, and wherein the blower is configured to receive the expanded and cooled air from the expander wheel and deliver the expanded and cooled air to an environment.